

ECOLOGY AND ECOSYSTEMS



NO LIVING THING exists in isolation. It interacts with other living things and with its physical surroundings. The study of these relationships is called ecology. Ecologists consider all the organisms that live in one area as an inter-dependent community. All plants and animals rely on, and influence, vital factors in their environment, such as the supply of nutrients, food, and water. A community and its environment is called an ecosystem.

Communities

Wildlife communities exist almost everywhere you look, on land, in rivers, and in the oceans. A typical community contains a mixture of plants, various animals that feed on them or hunt one another, and organisms that burrow through the soil debris below.

Trees offer shelter for animals, and food in the form of leaves, berries, seeds, and blossom.

Insects feeding on flowers help to pollinate them.

Habitats

The habitat of a species is the surroundings in which it lives, including the rocks, soils, water, and plants. Different habitats are suitable for different species and have a certain type of community.

Mice eat seeds, and are hunted by bigger animals.

Dense undergrowth provides shelter for small animals.

Most of the tadpoles that hatch out from the frog spawn will be food for other animals.

Rotting wood is home to fungi and invertebrates.

Snails feed on the leaves of plants and are food for some birds such as thrushes.

As ferns grow, they take nutrients from the soil.

Frogs live in both land and water habitats.

Biomes

The biggest ecological units are biomes, such as deserts, rainforests, and lakes, across which similar climatic and other conditions create similar ecosystems. The plants and animals may differ across a biome, but they make up the same sort of communities with the same ecological features.



Grassland

Grassland is normal in places where there is a long dry season. It can support lots of grazing animals, some preyed on by swift-running predators. The savannah of East Africa is one of the best-known areas of grassland.

Rainforests

In hot, humid climates, dense forests develop that are home to a huge variety of animal life. Tropical rainforests cover only 10 per cent of the Earth's land surface, but contain more than half of all animal and plant species.



Seashores

Battered by waves and flooded by tides, seashores have few plants other than seaweeds. Animals include shellfish, rockpool fish, and wading birds.



Deserts

Cloud-free, dry climates create deserts. These are home only to plants and animals that are able to cope with extremes of aridity and temperature.

Ecosystems

An ecosystem contains several different wildlife communities and their habitat. Ecologists use the term to mean all the complicated interactions that take place among living and non-living things in an area. The various components of the ecosystem include sunshine, water, nutrients in the soil, bacteria, plants, and animals.



Fresh water

Lake- and river-dwelling communities include floating or submerged plants, freshwater plankton, and fish. Different species live in different parts of a river or lake, depending on the conditions that they tolerate. This is Bow Lake in the Canadian Rockies.